ICT Level 1

Teaching Guide

Level 1 - Class 6

Level: Level 1

Class: Class 6

Duration: June - January (8 months)

Period: 40 minutes

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Curriculum Overview

ICT Level 1 is designed for Class 6 students in residential schools. The curriculum follows the 5E instructional model and integrates Singapore's Concrete-Pictorial-Abstract approach for effective learning.

Themes Covered

Chapter	Theme	Periods	Key Topics
1	Internet and ICT Environment	12	Introduction to ICT, Internet basics, Web browsers, GPS
2	Graphics in ICT	12	Painting/Drawing, Graphics software, Symmetric designs
3	Audio & Visual Communication	12	Communication types, Audio/Video conferencing, Audacity
4	Data Representation & Processing	12	Number systems, Data representation, Basic processing
5	Programming	12	Scratch programming, Basic blocks, Simple animations
6	Software Applications	12	Introduction to software, Types of software

Pedagogical Approach

5E Model Implementation

Phase	Duration	Activities
Engage	5 min	Real-world connection, regional metaphors, questions
Explore	10 min	Hands-on activity, Driver-Navigator pairs, Lab work
Explain	10 min	Live coding, Concept introduction, C-P-A approach
Elaborate	10 min	Extended activity, Parson's problems, Application
Evaluate	5 min	Formative check, Exit tickets, Self-assessment

International Pedagogies Used

- **Singapore C-P-A:** Concrete manipulatives -> Visual blocks -> Abstract code
- **UK NCCE:** Unplugged Computing, Live Coding, Parson's Problems
- **Driver-Navigator-Checker (Triad):** 3-person lab roles for 10 systems - Driver at keyboard, Navigator reads steps, Checker tests (NCCE pair programming adapted)
- **Station Rotation (50 students, 10 PCs, 40 min):** 25 in Lab (10× triad/dyad) + 25 in Library (Unplugged) - swap next lesson; roles 3+2 Hybrid when single room; librarian supervised (Christensen Institute)
- **Rural Scaffolding:** Regional metaphors (grain sorting, bus routes, hostel inventories)

Lesson Plan Structure (40 Minutes)

Time	Phase	Teacher Activity	Student Activity
0-5 min	Engage - Retrieval & Hook	Present real-world scenario using regional metaphor	Discuss, share observations
5-15 min	Explore - Guided Discovery	Guide hands-on exploration in pairs	Work in Driver-Navigator pairs, explore software
15-25 min	Explain - Modelling	Live coding demonstration, introduce concepts	Follow along, ask questions, take notes
25-35 min	Elaborate - Application	Assign extension activity or Parson's problem	Apply learning, solve problems independently
35-40 min	Evaluate - Hinge Q & Exit	Conduct formative assessment, assign homework	Complete exit ticket, receive dormitory task

Anatomy of a Lesson Plan (v2, August 2026)

Every lesson plan PDF now follows this fixed anatomy. Teachers should read top-to-bottom once before class; the plan is print-ready in black-and-white.

- **Header bar** - Lesson number, topic (italic), duration (40 min), and an archetype badge: CONCEPT (5E inquiry), LAB (Rosenshine I-do/We-do/You-do), or DEBUG (PRIMM Predict-Run-Investigate-Modify-Make). Chapters 2-3 are tagged LAB; chapters 5-6 DEBUG.
- **Success Criteria** - ‘I can…’ statements derived from objectives; used again during Evaluate for self-assessment ticks.
- **Resources Needed** - Placed at the TOP so materials can be gathered before class. Split mentally into Lab Kit vs Library Paper Pack for the 25/25 station rotation.
- **Key Vocabulary** - Four terms with one-line definitions for EAL learners; pre-teach during Engage.
- **Five phases** (5/10/10/10/5) - Each phase shows TEACHER and STUDENTS blocks side-by-side by left rule colour (blue/green).
- **Differentiation** - Four tiers: Approaching / On Level / Advanced / SEND.
- **Formative Assessment** - Method plus the Computational Thinking skill assessed (maps to the CT rubric in assessments).
- **Hinge Question** - Embedded in Evaluate; if fewer than 80% answer correctly, reteach 3 minutes before Elaborate.
- **Dormitory Task** - Interleaved retrieval: 3 facts from today, 2 connections to previous chapter, 1 question, plus one focused task and a weekly spiral recall.
- **Colophon** - Author, license (CC BY-SA 4.0), version.

Assessment Framework

Assessment Components (Per Class)

Assessment Type	Count	Marks Each	Total
Formative (FA-1 to FA-4)	4	30	120
Unit Tests	4	10	40
Summative (SA-1, SA-2)	2	40	80
Practical Lab Exams	2	20	40
Grand Total	**12**		**280**

Resources Required

Hardware

- Computer lab with 1 PC per 2 students (minimum)
- Internet connection (2 Mbps minimum)
- Projector for demonstrations
- Speakers and headphones

Software (Free/Open Source)

- Operating System: Ubuntu Linux or Windows
- Office Suite: LibreOffice / Google Workspace
- Graphics: GIMP, Tux Paint
- Audio: Audacity
- Programming: Scratch
- Browser: Firefox / Chromium

Dormitory Tasks (Zero Lab Access)

Lesson plans now embed interleaved 3-2-1 retrieval as the standard dormitory task format. For residential schools with limited lab time, these paper-and-pencil tasks can be assigned for supervised evening study:

- **Code Dry-Running:** Trace Scratch programs on paper
- **Variable State Tables:** Track variable values through program execution
- **Flowchart Drawing:** Create flowcharts for algorithms
- **Binary Conversion:** Practice decimal to binary conversions
- **Keyboard Layout:** Practice typing without computer
- **Software Matching:** Match software to their purposes

Capstone Projects

Project 1: School Environment Monitor

Integration: Science + ICT

Description: Students create a Scratch program that simulates monitoring temperature, humidity, and weather conditions in the school campus.

Skills: Data representation, variables, loops, conditionals

Project 2: Hostel Store Inventory

Integration: Mathematics + ICT

Description: Students create a spreadsheet to track hostel store items (rice, dal, sugar) with quantity, rate, and total calculations.

Skills: Spreadsheet formulas, data entry, basic calculations

Unplugged Activities Library

These activities teach computational thinking without any computer access. Use them in regular classrooms, during lab shortages, or as warm-up exercises.

Activity 1: Human Robot (Ch 5 - Programming)

Duration: 15 minutes | **Materials:** None

Steps:

1. One student is the “robot”, another is the “programmer”
2. The programmer gives step-by-step commands: “Move forward 3 steps”, “Turn left”, “Pick up the pencil”
3. The robot follows ONLY exact instructions (cannot interpret vague commands)
4. Switch roles. Discuss: What happens when instructions are unclear?

CT Skill: Algorithm design, precision in communication

Activity 2: Packet Relay (Ch 1 - Internet)

Duration: 20 minutes | **Materials:** Paper slips, tape

Steps:

1. Students stand in a line (the “network”)
2. Each student is a “router” who receives a message (paper slip) and passes it forward
3. If a student receives a damaged message (crumpled slip), they must request retransmission
4. Time how long the message takes to travel. Discuss: How does the internet work?

CT Skill: Decomposition, understanding networks

Activity 3: Pixel Art Grid (Ch 2 - Graphics)

Duration: 15 minutes | **Materials:** Graph paper, colored pencils

Steps:

1. Give students a small grid (8x8) and ask them to fill squares to create a simple image
2. Use only 4 colors. Discuss: How is this like a digital image?
3. Count total pixels used. What happens if you use a bigger grid (16x16)?

CT Skill: Abstraction, pattern recognition

Activity 4: Sorting Race (Ch 4 - Data)

Duration: 15 minutes | **Materials:** 30 mixed beans/grains, paper cups

Steps:

1. Give each pair a cup of mixed grains (rice, dal, mustard seeds)
2. Race to sort them into separate groups
3. Try different sorting strategies: one-by-one vs. grouping by type first
4. Discuss: Which strategy was faster? Why?

CT Skill: Pattern recognition, algorithm efficiency

Activity 5: Debugging Challenge (Ch 5 - Programming)

Duration: 10 minutes | **Materials:** Printed “code” on paper

Steps:

1. Give students a simple Scratch-like program with 3 intentional errors
2. Students must find and fix each error
3. First to find all errors wins
4. Discuss: How do you find mistakes in your own work?

CT Skill: Debugging, logical reasoning

Activity 6: Story Sequencing (Ch 3 - Audio/Visual)

Duration: 15 minutes | **Materials:** Printed story cards (cut up)

Steps:

1. Give groups 8-10 cards with steps of a story (e.g., “How to make Ragi Mudde”)
2. Cards are shuffled. Groups must arrange them in correct order
3. Some cards may be unnecessary. Identify which ones to remove
4. Discuss: How is this like editing a video?

CT Skill: Sequencing, abstraction

Cross-Curricular Integration Map

Mathematics Connections

- **Ch 1:** IP addresses use numbers (decimal/binary conversion)
- **Ch 2:** Pixel coordinates are geometry (x, y positions)
- **Ch 3:** Audio sampling rates involve measurement
- **Ch 4:** Data representation uses statistics (mean, median, mode)
- **Ch 5:** Loops use counters and variables (algebra)
- **Ch 6:** File sizes use units (bytes, kilobytes)

Science Connections

- **Ch 1:** Network signals travel as electromagnetic waves (Physics)

- **Ch 2:** Light and color mixing (RGB model - Physics/Chemistry)
- **Ch 3:** Sound waves and frequency (Physics)
- **Ch 4:** Data collection follows scientific method
- **Ch 5:** Algorithms model real-world processes (Biology, Chemistry)
- **Ch 6:** Software runs on hardware (Physics of electronics)

Language Connections

- **Ch 1:** Writing clear instructions = algorithm design
- **Ch 2:** Visual storytelling through graphics
- **Ch 3:** Audio narration and script writing
- **Ch 4:** Data description and report writing
- **Ch 5:** Program comments = technical writing
- **Ch 6:** Documentation and user guides

Digital Citizenship Module

Class 6: Being Safe Online

Duration: 2-3 lessons (can be integrated into Ch 1)

Lesson: Passwords and Privacy

- Create strong passwords using a formula: Favorite food + birth year + special character
- Practice: “Ragi2014!” vs “password123”
- Rule: Never share passwords with anyone except parents/teachers

Lesson: Identifying Fake News

- Show examples of fake WhatsApp forwards (use local language examples)
- Check: Is this from a known source? Does it seem too shocking?
- Rule: “When in doubt, don’t share”

Lesson: Screen Time Balance

- Suggest: 30 minutes screen time, then 10 minutes looking at green trees
- Create a “Screen Time Chart” in their notebook
- Rule: Eyes need rest, body needs movement

Class 7: Being Responsible Online

Duration: 2-3 lessons (can be integrated into Ch 6)

Lesson: Cyberbullying Prevention

- What to do if someone sends mean messages: Tell a trusted adult
- What NOT to do: Don’t reply, don’t forward, don’t keep it secret
- Create a “Help Card” with contact numbers

Lesson: Digital Footprint

- Everything you post online stays forever
- Exercise: Google yourself (or your school). What do you find?
- Rule: “Don’t post anything you wouldn’t want your teacher to see”